PRM Scores Are Not Pruning Policies: A Counterfactual Study of Reasoning-Step Removability

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Abstract

Process reward models (PRMs) score the local correctness of reasoning steps, and reasoning systems increasingly reuse those scores to prune "bad" steps for efficiency or accuracy. We ask whether step-level correctness signals actually identify *removable* steps. We introduce a counterfactual intervention protocol—control, deletion, position-and-length-matched *placebo* deletion, and semantic-preserving *paraphrase*—and apply it to 600 rating-balanced PRM800K steps (over 10,000 regenerated continuations), with replications on a second generator and on 300 ProcessBench steps from twelve source models. Deletion raises accuracy by $+7.2$ points on average, but the gain concentrates almost entirely in steps that are both low-rated and semantically harmful ($+23.3$ pp; phi-4: $+33.7$ pp; external first-error steps: $+24.8$ pp). Controls decompose the mechanism: placebo deletion accounts for two fifths of the anchor-group gain (generic restart), while paraphrasing the erroneous step—preserving its wrong meaning—performs like keeping it, so the residual $+14$–19 pp is attributable to the step's *content*: a negative anchor. Crucially, no correctness signal identifies safe deletions: the dataset rating ranks deletion danger below chance (AUROC 0.42), a deployed PRM checkpoint and prompted PRM judges do no better, step entropy is uninformative, and under a 1% harm budget no score-threshold policy can delete a single step—while a five-feature predictor deletes a third of the cohort, and a single probe-and-rollback generation cuts harmed steps from 12% to 3%. PRM scores are evidence for *ranking* pruning candidates, not licenses to delete; safe pruning needs counterfactual validation. We release the audited dataset, pipeline, and benchmark.

1 Introduction

Process reward models assign scores to individual steps of a reasoning trajectory and have become a standard tool for verifying and steering large language model (LLM) reasoning (Lightman et al., 2024; Uesato et al., 2022; Wang et al., 2024). As long chains of thought grow expensive, a tempting reuse of these scores has emerged: treat low-scoring steps as disposable, and prune them to shorten context, save tokens, or even repair a derailed trajectory (Choi et al., 2025; Li et al., 2025). This reuse embeds a silent assumption: that a step's *correctness score* tells us what happens to downstream reasoning if the step is *removed*.

This assumption conflates several distinct properties of a reasoning step. A step can be locally correct yet contribute nothing that later steps use; it can be locally incorrect yet occur after the trajectory is already doomed, so removing it changes nothing; and it can be incorrect in a way that actively *anchors* the continuation to a bad path, so removing it rescues the run. Whether deletion helps is a *counterfactual* property of the step–generator pair, not a syntactic property of the step in isolation.

We study this gap directly. Starting from PRM800K (Lightman et al., 2024), we sample 600 target steps balanced across human ratings ($-1/0/+1$), annotate each step's semantic contribution (Essential / Redundant / Harmful), and measure what actually happens when a fixed generator (Qwen3-8B; Qwen Team, 2025) continues the trajectory with and without the step, four runs per condition, 2,400 paired outcomes. Because deleting *any* step also forces the generator to re-plan—which can itself change outcomes—we additionally run a position-and-length-matched *placebo* deletion for 511 of the 600 steps (1,514 runs), letting us decompose raw deletion effects into a target-specific semantic component and a generic restart component. An LLM judge scores final answers; a 200-output human audit bounds judge error and a full label-substitution analysis bounds its effect on every headline number (≤ 0.13 pp).

Three findings emerge. **First, correctness and**

contribution are correlated but far from interchangeable (Cramér's $V = 0.53$; 34.5% of steps off the expected diagonal), and deletion outcomes track the *interaction*: only steps that are both low-rated and Harmful yield large reliable gains (+23.3 pp), while every other rating or type group is statistically indistinguishable from zero. The pattern replicates on a second generator from a different pretraining pipeline (phi-4: anchor effect +33.7 pp) and on external ProcessBench trajectories from twelve source models. **Second, deletion gains are not all semantic—but the anchor component is.** The matched placebo reveals a +10.0 pp generic "restart" component inside the low-rated Harmful group, and a semantic-preserving paraphrase control pins down the remainder: rewording the erroneous step leaves accuracy at control level while deleting it gains +19.2 pp over the paraphrase—erroneous *content*, not surface form or token count, drags continuations toward failure (a *negative anchor*). **Third, scores are not policies.** Under a 1% harm budget, *no* threshold on the PRM800K rating, the semantic type, or the continuous score of a deployed PRM checkpoint can delete a single step, while a five-feature logistic predictor deletes 200 of 600 steps at the same budget and an oracle deletes 530. The information gap, not the intervention, is the bottleneck—and a single probe-and-rollback generation recovers most of the missing safety (harmed steps 12.2% → 3.0%).

Our contributions: (i) a conceptual separation of step correctness, semantic contribution, and retrospective removability, with pre-registered estimands; (ii) a reusable counterfactual protocol—control / target deletion / matched placebo / semantic-preserving paraphrase—with an audited automatic evaluator; (iii) an empirical characterization of *negative reasoning anchors*, replicated across two generators and an external multi-model benchmark; (iv) a signal audit spanning dataset ratings, deployed PRM scores, uncertainty, and counterfactual importance; and (v) a risk–coverage and rollback benchmark showing correctness thresholds are unusable as deletion policies under realistic harm budgets, while cheap counterfactual validation is not. We release all data, code, and audits.

2 Related Work

Process supervision and PRMs. Step-level supervision improves credit assignment over outcome-only rewards (Uesato et al., 2022; Lightman et al., 2024; Cobbe et al., 2021). Math-Shepherd derives step labels from rollout success probabilities (Wang et al., 2024); Setlur et al. (2025) argue for rewarding *progress* (prospective advantage) rather than local correctness; generative PRMs verify steps with explicit reasoning (Khalifa et al., 2025). Benchmarks probe what PRMs detect: ProcessBench locates first errors (Zheng et al., 2025), and PRMBench separates fine-grained error dimensions including redundancy (Song et al., 2025). None of these measure what happens when a step is *removed*; our protocol supplies exactly that counterfactual layer, and our results show that correctness-oriented signals transfer poorly to it.

Reasoning compression and pruning. Token- and step-pruning methods compress chains of thought using redundancy heuristics, entropy, or self-evaluation (Choi et al., 2025; Li et al., 2025), and probing studies ask whether models internally encode token importance (Singh and Hakkani-Tür, 2025). These works optimize for preserving the answer of an already-correct trajectory; our study additionally quantifies the *repair* regime (deleting misleading content from failing trajectories) and shows the two regimes respond to different signals.

Deletion as intervention. Ablation-style causal analyses of reasoning typically mask content at the token level or measure answer-probability shifts. Our design differs in running full *regenerated* continuations—the deployment-relevant intervention—and in controlling the regeneration itself with a matched placebo, which we show is essential: without it, restart effects of up to +10 pp inside key subgroups would be misattributed to step semantics.

3 Constructs and Estimands

We distinguish four properties of a step s_t in trajectory h: (1) *local correctness*, the PRM800K rating in $\{-1, 0, 1\}$; (2) *semantic contribution*, whether the content of s_t is needed by (Essential), irrelevant to (Redundant), or misleading for (Harmful) downstream reasoning; (3) *prospective advantage*, the change in success probability from *taking* the step (Setlur et al., 2025); (4) *retrospective removability*, the change in success probability from *deleting* the step after the fact and regenerating. This paper measures (4) and its relation to (1) and (2).

For step i with paired runs $j = 1{:}K$ ($K{=}4$), let $Y^c_{ij}, Y^t_{ij}, Y^p_{ij} \in \{0, 1\}$ be final-answer correctness under **C**ontrol (step kept), **T**arget deletion, and **P**lacebo deletion (a different step of matched position and length deleted). Pre-registered estimands:

$$\Delta_i^{\text{tgt}} = \bar{Y}_i^t - \bar{Y}_i^c, \qquad \Delta_i^{\text{plc}} = \bar{Y}_i^p - \bar{Y}_i^c,$$
$$\Delta_i^{\text{sem}} = \Delta_i^{\text{tgt}} - \Delta_i^{\text{plc}}, \tag{1}$$

plus run-level transition events *danger* ($Y^c{=}1 \rightarrow Y^t{=}0$) and *benefit* ($Y^c{=}0 \rightarrow Y^t{=}1$). All confidence intervals are 95% percentile intervals from a 5,000-replicate cluster bootstrap resampling *target steps* (the unit at which interventions are randomized); fixed seeds are released.

4 Data and Intervention Protocol

Step cohort. From PRM800K phase-2 test trajectories (Lightman et al., 2024) over MATH problems (Hendrycks et al., 2021) we sample 600 target steps, 200 per rating, balanced over early/middle/late positions ($201/201/198$). Semantic contribution was annotated twice: an initial AI-assisted pass, and a human-calibrated pass in which the authors relabeled with a few-shot calibrated rubric; the two label sets disagree on 164/600 steps (27.3%), which we treat as an explicit reliability bound and revisit in the Limitations. Under the calibrated labels the cohort contains 198 Essential, 199 Redundant, and 203 Harmful steps. Rating and type are associated (Cramér's $V = 0.528$) with 65.5% of steps on the expected diagonal ($+1\leftrightarrow$Essential, $0\leftrightarrow$Redundant, $-1\leftrightarrow$Harmful)—correlated, but with a third of the cohort off-diagonal (Figure 1).

Interventions. For each step we construct prompts containing the problem plus the trajectory prefix through step t (Control) or through step t with s_t excised (Target deletion), and let Qwen3-8B (temperature 0.7, top-p 0.8) regenerate the continuation; four runs per condition. For the placebo condition we delete instead a different step from the same prefix whose token length is within $\pm 20\%$ of the target's, choosing the closest available position; 511/600 steps admit such a match (1,514 placebo runs), and the 89 unmatched steps differ from matched ones on no outcome-relevant covariate (accuracy, harm, and recovery differences all cross zero; maximum position SMD 0.298), differing only in continuation-length change.

Outcome evaluation and audit. An LLM judge marks final-answer correctness against ground truth. We audited 200 stratified outputs with human adjudication: agreement 93.5% (F1 0.931), pair-transition agreement 91.7%, and a maximum per-condition bias of 2.3 pp—below the pre-set gates (90% / 5 pp). Substituting all 200 adjudicated labels into the frozen tables moves the overall target effect by -0.13 pp and the pure semantic effect by -0.05 pp; no conclusion in this paper is sensitive to judge noise at this scale (Table 7).

Replication and mechanism cohorts. Four extensions reuse the identical prompt protocol. (i) *Cross-generator*: 300 steps (all 100 available rating $-1\times$Harmful anchors, 100 rating-0, 100 rating-1; 103 stably-correct; positions balanced) rerun with **phi-4** (14B; a different pretraining pipeline from Qwen), 3 runs per condition, reusing the frozen matched-placebo indices.[1] (ii) *Semantic-preserving paraphrase* (condition C3): for a 240-step strong-control subset (80 anchors / 80 stably-correct / 80 rating-$0/1$ fragile), an independent model (Gemini 2.5 Flash) paraphrases the target step preserving its exact meaning *including errors*; 221/240 paraphrases passed automatic validity checks (7.9% excluded) and each ran 4 continuations, alongside 4 fresh matched-placebo runs (C2). (iii) *Deployed PRM scoring*: three PRMs score all 600 steps—the trained checkpoint Qwen2.5-Math-PRM-7B (full-trajectory, token-level head) and two prompted LLM-judge PRMs (Gemini 2.5 Flash, phi-4). (iv) *External validation*: 300 ProcessBench steps (Zheng et al., 2025) spanning GSM8K, MATH, OlympiadBench, and Omni-MATH trajectories from twelve source models (Section 7).

5 Full-Cohort Deletion Results

Deleting target steps helps on average (+7.21 pp, Table 1), but the average is misleading: the entire reliable gain sits in the rating $= -1$ column and the Harmful row, peaking at +23.31 pp for their intersection. Correct steps (rating +1) and Essential steps show near-zero *average* effects—yet not because deletion is safe there. Danger is substantial everywhere: across the cohort, 15.1% (CI $[11.9, 18.4]$) of control-correct runs flip to wrong under deletion, and even within rating $= -1$ the danger rate is 14.9%. The near-zero group means arise from offsetting harm and benefit, not from inertness.

[1]The plan's nominal second generator (DeepSeek-R1-Distill) was unavailable or exhausted its token budget in reasoning without emitting parseable output; phi-4 matched the 7–14B budget and the no-thinking protocol. All 2,691 continuations parsed.

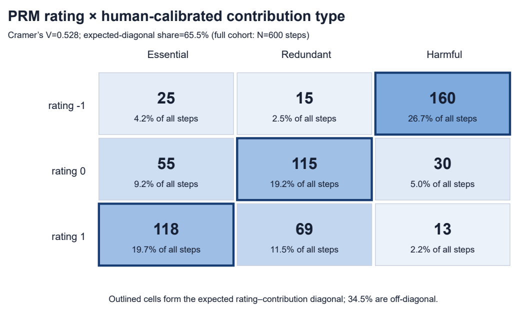

Figure 1: PRM800K rating vs. human-calibrated semantic contribution for the 600-step cohort. The association is strong (Cramér's $V = 0.528$) but 34.5% of steps are off-diagonal: local correctness does not determine contribution.

Group	Steps	Δ^{tgt} (pp)	95% CI
Overall	600	$+7.21$	$[+3.88, +10.58]$
rating $= -1$	200	$+22.00$	$[+15.41, +28.42]$
rating $= 0$	200	$+0.25$	$[-5.41, +5.90]$
rating $= +1$	200	-0.63	$[-5.54, +4.21]$
Essential	165	-1.21	$[-7.24, +4.78]$
Redundant	201	$+1.12$	$[-3.87, +6.06]$
Harmful	234	$+18.38$	$[+12.28, +24.44]$
$-1\times$Harmful	178	$+23.31$	$[+16.57, +30.24]$

Table 1: Target-deletion effects on final-answer accuracy (600 steps, 2,400 paired runs, cluster bootstrap over steps). Gains concentrate in low-rated Harmful steps; all other rating and type margins cross zero.

Trajectory state dominates raw effects. Stratifying by control stability makes the state-dependence explicit (Figure 2): steps whose four control runs are all wrong gain $+33.4$ pp from deletion (nothing to lose, 299 rescued runs), while steps whose control runs are all correct *lose* 12.1 pp (CI $[-15.5, -8.9]$; 144 destroyed runs, zero possible gains). A deployment policy never observes these counterfactual states, which is exactly why raw deletion gains overstate what a policy can capture (Section 9).

Step-level stability. With four paired runs per step, 377/600 steps never change outcome, 99 are strongly beneficial (gain in ≥ 3 runs), 57 strongly harmful, and the rest weak or mixed. Of the 178 low-rated Harmful steps, 51 (29%) are strongly beneficial to delete—$2.5\times$ the rate in the rest of the cohort (11%), but far from a guarantee.

Cross-generator replication. The pattern is not a Qwen3-8B artifact. Rerunning 300 steps with

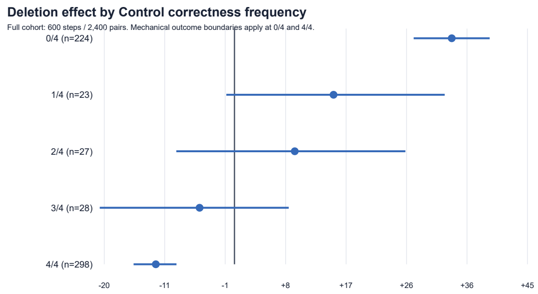

Figure 2: Deletion effect by control-run stability. Gains exist only where control already fails; stably-correct trajectories are harmed (-12.1 pp).

Group	Δ^{tgt}	Δ^{plc}	Δ^{sem}	CI(Δ^{sem})
Overall (511)	$+7.97$	-0.57	$+8.55$	$[+4.83, +12.17]$
rating $= -1$ (169)	$+22.78$	$+9.47$	$+13.31$	$[+6.71, +19.53]$
rating $= 0$ (176)	$+0.99$	-5.63	$+6.63$	$[+0.19, +13.21]$
rating $= +1$ (166)	$+0.30$	-5.42	$+5.72$	$[-0.05, +11.40]$
Harmful (201)	$+18.41$	$+3.57$	$+14.84$	$[+8.83, +20.61]$
$-1\times$Harmful (151)	$+23.84$	$+9.99$	$+13.85$	$[+6.95, +20.64]$

Table 2: Placebo decomposition on the 511-step matched cohort (effects in pp). Deleting a *matched innocuous step* already yields $+10.0$ pp inside the low-rated Harmful group: 42% of the raw deletion gain there is restart, not semantics.

phi-4 (3 runs per condition) reproduces every qualitative conclusion: overall target effect $+12.4$ pp (CI $[+8.3, +16.8]$), anchor-group target effect $+33.7$ pp (CI $[+25.7, +42.0]$) with a pure semantic component of $+18.3$ pp (CI $[+10.7, +26.0]$), and rating-0/rating-1 target effects indistinguishable from zero ($+1.7$ and $+2.0$ pp). Effect *magnitudes* are generator-dependent (phi-4's anchor effects are larger, consistent with weaker recovery from bad prefixes), and step-level effect signs agree for 65% of the 80 steps with nonzero effects under both generators—well above chance but far from deterministic. Removability generalizes as a population-level phenomenon while remaining partly a property of the step–generator pair.

6 Mechanism: Anchors vs. Restarts

Does deletion help because the *content* of the step was harmful, or because *any* deletion lets the generator restart and resample? Table 2 separates the two. Overall, placebo deletion is inert (-0.57 pp), and the pure semantic effect ($+8.55$ pp) essentially equals the raw effect. But inside the low-rated groups the placebo is *not* inert: deleting a matched innocuous step from a trajectory that contains a low-rated Harmful step yields $+9.99$ pp by itself. Intuitively, these trajectories are so fragile

Contrast (pp)	Anchor	Stable-corr.	Fragile-neutral
C1−C0 (raw deletion)	+20.3	−11.9	+24.1
C1−C2 (vs. placebo)	**+15.6**	+16.8	**+0.0**
C3−C0 (paraphrase)	**+1.7**	−8.3	+20.9
C1−C3 (vs. paraphrase)	**+19.2**	−2.7	+4.1

Table 3: Four-condition strong controls (240 steps × 4 runs; 80 per group; paraphrase available for 221). Anchor group: paraphrase ≈ control while deletion beats both—the *content* is the anchor. Fragile-neutral group: deletion = placebo—pure restart. Stably-correct group: even meaning-preserving rewording hurts (−8.3 pp).

that any perturbation that forces re-planning has positive expected value. The target-specific component that survives subtraction—+13.85 pp (CI [+6.95, +20.64]) for rating −1×Harmful—is our headline evidence for *negative anchors*: erroneous step content that actively pulls regenerated continuations toward failure, over and above the restart benefit.

Two methodological corollaries. First, deletion studies without matched placebos conflate these components; in our data the conflation is worth up to 10 pp precisely in the subgroups where deletion looks most attractive. Second, restart effects are heterogeneous (negative for rating 0/+1 groups, strongly positive for fragile ones), so a single global "restart correction" would also be wrong; matching must be per-step.

Paraphrase triangulation. The placebo isolates *that* the target matters; a semantic-preserving paraphrase isolates *why* (Table 3). In the anchor group, replacing the erroneous step with an independent-model rewording that preserves its (wrong) meaning leaves accuracy at control level (+1.7 pp, CI [−6.2, +9.9]), while deleting it gains +19.2 pp over the paraphrase (CI [+8.2, +30.1]): the harm travels with the *semantics*, not the surface form—the definitive signature of a negative anchor. The fragile-neutral group shows the mirror image: a large raw gain (+24.1 pp) that placebo deletion matches exactly (C1−C2 = 0.0 pp), i.e. pure restart. And stably-correct steps are harmed even by faithful rewording (−8.3 pp, CI [−14.7, −3.0]), showing that correct reasoning chains are sensitive to perturbations that preserve meaning—deletion is not the only risky edit. A blinded human fidelity audit backs the C3 premise: four annotators rated 140 sampled paraphrases (original and rewording side by side, groups hidden), judging 97.1% faithful or minimally deviating; in the anchor group, zero of 46 audited paraphrases corrected the original error,

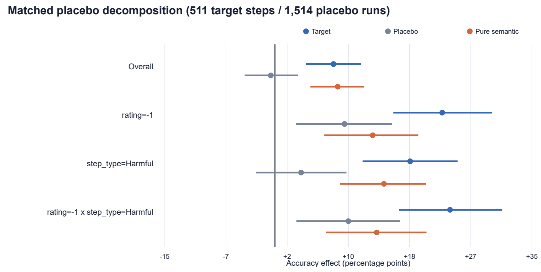

Figure 3: Target, placebo, and pure-semantic effects with 95% cluster-bootstrap CIs, matched 511-step cohort.

so C3 does not silently repair the step it rewords.

Qualitative families. Eight fully human-verified cases (78 audited continuations) illustrate the mechanism space: 3 negative anchors (deletion removes a wrong commitment and continuations recover), 2 generic restarts (placebo helps as much as target deletion), 2 stably-correct steps destroyed by deletion, and 1 high-rated Redundant step whose deletion flips outcomes unpredictably. We release all cases with full continuations.

7 External Validation on ProcessBench

Does any of this survive outside PRM800K trajectories and outside human step ratings? We sample 300 ProcessBench steps (Zheng et al., 2025)—152 human-annotated *first-error* steps and 148 *locally-correct* steps, balanced across GSM8K, MATH, OlympiadBench, and Omni-MATH (75 each) and drawn from trajectories written by twelve different source models—and run the identical protocol (3 runs per condition; 204/300 steps admit a length-matched placebo; gold answers matched from source datasets for 99.5% of records).

The core dissociation transfers. Deleting first-error steps yields +24.8 pp (CI [+16.9, +32.7]), with a placebo-corrected semantic component of +19.4 pp (CI [+10.2, +28.3]); deleting locally-correct steps yields −2.0 pp (CI [−8.3, +4.1])—erroneous steps in other models' reasoning are negative anchors too, and locally-correct steps are not usefully removable even though they are also not reliably *dangerous* on average. The overall semantic effect (+8.2 pp, CI [+2.1, +14.2]) is strikingly close to the PRM800K cohort's +8.6 pp. One boundary emerges: the semantic component is significant on GSM8K (+17.5 pp) and MATH (+13.9 pp) but attenuates on OlympiadBench (+6.8 pp, CI crosses zero) and vanishes on Omni-MATH (−3.8 pp), where placebo deletion

Model (features)	Danger AUPRC	Benefit AUPRC
Base rate	0.151	0.360
A: rating only	0.139	0.432
B: type only	0.208	0.407
C: rating + type	0.140	0.380
D: static context	0.160	0.412
E: + trajectory state	**0.263**	**0.478**

Table 4: Out-of-fold prediction of run-level danger (base rate 15.1%) and benefit (base rate 36.0%). The rating is *below chance* for danger (AUROC 0.42, CI [0.36, 0.49]); only trajectory-state features (model E, requiring extra runs) achieve a meaningful margin (danger AUPRC +0.10 over D, CI [+0.05, +0.16]).

	Danger		Benefit	
Signal (step-level)	AUROC	AUPRC	AUROC	AUPRC
Base rate	—	0.15	—	0.24
PRM800K rating	0.56	0.17	0.61	0.34
Qwen2.5-Math-PRM-7B	0.57	0.18	0.61	**0.37**
LLM-judge PRM (Gemini)	0.53	0.16	0.51	0.25
LLM-judge PRM (phi-4)	0.49	0.14	0.48	0.23
PRM disagreement	0.57	0.18	**0.62**	**0.38**
Step entropy	0.46	0.15	0.48	0.24
Step NLL	0.50	0.15	0.49	0.22
Masked answer-prob drop	0.56	0.19	0.57	0.30

Table 5: Step-level signals vs. deletion outcomes on the identical 600-step cohort (danger/benefit = any correct-to-wrong / wrong-to-correct transition; step base rates 0.15/0.24; signals oriented so higher warns against / flags for deletion respectively). Benefit AUPRCs reach 0.37–0.38 over a 0.24 base—a real lift—while the best danger AUPRC (0.19) barely exceeds its 0.15 base. Uncertainty signals (entropy, NLL) are at chance throughout.

alone trends positive (+9.4 pp): on problems near the generator's competence ceiling, rescues come from restarting rather than from removing specific bad content. Negative anchors are a phenomenon of *recoverable* trajectories.

8 Do Correctness Signals Predict Removability?

If PRM-style signals suffice for pruning, they should predict the two events a policy cares about: *danger* (destroying a correct run) and *benefit* (rescuing a wrong one). We evaluate step-level signals with 5-fold cross-validation grouped by problem (no problem appears in both train and test), L2-regularized logistic models, and cluster-bootstrap CIs (Table 4).

The asymmetry matters. Signals rank *benefit* moderately well: low rating and Harmful type genuinely enrich for rescuable runs. But for *danger*—the safety-critical direction—the rating alone is anti-informative (AUROC 0.422): among runs whose control is correct, low-rated steps are *not* safer to delete, and conditioning on the rating actively misleads. Only features that peek at trajectory state (control stability; effectively extra compute) move danger prediction appreciably. Cheap local correctness is calibrated for "was this step wrong?", not for "will removing it break the run?".

Deployed PRMs, uncertainty, and counterfactual signals. Because "PRM scores" in deployment are continuous model outputs rather than dataset ratings, we scored the identical 600 steps with a trained PRM checkpoint (Qwen2.5-Math-PRM-7B) and two prompted LLM-judge PRMs, plus step entropy/NLL under teacher forcing and a masked answer-probability drop $I_{\text{mask}} = \log p(y^\star \mid h) - \log p(y^\star \mid h \setminus s_t)$ (Table 5). The trained PRM tracks the human rating almost exactly (danger AUROC 0.57 vs. 0.56)—so the construct gap is not an artifact of auditing labels instead of scores. Token-level uncertainty carries no removability information at all. Two signals stand out modestly: disagreement between independent PRMs (the best benefit ranker, AUPRC 0.38), supporting uncertainty-based *abstention*, and the counterfactual mask drop, the only answer-conditioned signal—informative but requiring the gold answer, hence useful for offline auditing rather than deployment.

9 From Scores to Policies: Risk–Coverage

We convert the audit into a deployment question: if a system deletes the top-k steps ranked by a signal, what net accuracy does it gain, and how many previously-correct runs does it destroy? We evaluate seven policies offline on the frozen cohort: random; threshold policies (rating $= -1$; Harmful-only; their intersection); the model-D and model-E predictors of Section 8 (ranked by out-of-fold $P(\text{benefit}) - P(\text{danger})$, with base-rate imputation so the static policy cannot exploit prediction-availability leakage); and an oracle ranked by observed effect. *Harm rate* is correct-to-wrong transitions per deleted run.

Table 6 is the operational core of the paper. At its natural operating point (all 200 rating $= -1$ steps), the rating-threshold policy gains +7.3 pp net accuracy but destroys correct runs at a 4.6% rate—tolerable only under the loosest budget. Under a 1% budget, *every* rating/type threshold col-

Policy	Harm budget 1%	3%	5%
Random	0	0	0
rating $= -1$	0	50	210
Harmful-only	0	10	180
rating $= -1 \times$Harmful	0	10	240
Predictor (static, D)	**200**	240	380
Predictor (state, E)	**210**	250	280
Oracle upper bound	530	550	570

Table 6: Maximum deletable steps (of 600) under fixed harm budgets, where harm is correct-to-wrong transitions per deleted run. No correctness-threshold policy deletes *anything* within a 1% budget; a five-feature CV predictor deletes a third of the cohort. The oracle shows the gap is informational, not intrinsic.

lapses to zero coverage, because the thresholds cannot separate fragile-but-rescuable trajectories from stable-correct ones. The learned predictors, using only cheap static features plus (for E) trajectory-state estimates, delete 200–210 steps at 1% harm with +11 pp net gain. The oracle deletes 530. The information needed for safe pruning exists in the outcome structure; PRM-style correctness scores simply do not carry it.

Deployed-PRM thresholds fare no better. Ranking deletions by the trained PRM's continuous score yields 20 deletable steps at the 1% budget (200 at 5%); the LLM-judge PRM yields zero at 1% and 3%, and a two-PRM ensemble that abstains on disagreement reaches only 100 steps even at 5%. The failure of Table 6's top rows is a property of correctness signals as a class, not of the PRM800K rating in particular.

Rollback: one probe run buys most of the safety. Finally, we simulate the cheapest counterfactual validator: generate *one* probe continuation after a candidate deletion, roll back unless the probe answers correctly, and evaluate on held-out runs. Applied to all 600 steps, this cuts the share of harmed steps from 12.2% to 3.0% while *raising* the mean per-step gain from +7.2 to +11.5 pp (on anchors: 8.1% $\rightarrow$ 1.9%), at a cost of one extra generation per candidate and a 63% acceptance rate. Validation is not merely necessary (Sections 8–9); it is also cheap relative to the value it protects.

10 Discussion

What PRM scores are for. Our results do not indict PRMs at their trained task: low ratings do flag wrong steps, and wrong steps are where deletion *benefit* concentrates. The failure is direction-specific: scores carry almost no information about deletion *danger*, which is governed by trajectory state and generator recoverability. The safe role for a PRM score in a pruning system is candidate ranking—proposing which steps to *consider* deleting—followed by counterfactual validation (regenerate-and-check with rollback) before commitment. Score thresholding alone, at any threshold we can construct, either deletes nothing or accepts multi-point harm rates.

Negative anchors. The surviving target-specific effect in low-rated Harmful steps supports a mechanism in which erroneous content behaves as an attractor for regeneration: continuations conditioned on the anchor commit to its error, while continuations from the cleaned prefix explore correct paths. That placebo deletion alone recovers +10 pp in the same trajectories shows fragility and anchoring coexist—both "the trajectory was rescuable by any shake-up" and "this step specifically was holding it down" are true, in roughly 40/60 proportion.

Toward deployable pruning. The oracle–policy gap (530 vs. 200 steps at 1% harm) quantifies the headroom for better removability signals—and Section 8 shows the obvious candidates do not close it: deployed PRM scores replicate the rating's failure, uncertainty signals are at chance, and the informative signals (mask drop, PRM disagreement, trajectory state) all cost extra computation or gold answers. The constructive path our results support is architectural rather than signal-side: use correctness scores to *nominate* candidates, spend one probe generation to *validate*, and roll back on failure—a loop that already recovers most of the oracle's safety at bounded cost.

11 Conclusion

Across 600 audited PRM800K steps, two generators, four external datasets, and over 13,000 regenerated continuations, step correctness and step removability dissociate: deletion gains concentrate in low-rated *Harmful* steps, part of that gain is a restart artifact that only a matched placebo can remove, the remainder travels with the step's erroneous content rather than its surface form, and no correctness signal—dataset rating, deployed PRM score, or uncertainty—ranks deletion danger usefully above chance. PRM scores are useful evidence about steps; they are not pruning policies.

Systems that delete reasoning content should nominate with scores, validate counterfactually, and budget harm explicitly—and benchmarks for PRMs should test not only whether models find errors, but whether their scores license the interventions built on top of them.

Limitations

Generator scope. Primary results use Qwen3-8B; the replication adds phi-4 (300 steps, 3 runs). Both are mid-size open models run without extended thinking; effect magnitudes differ across the two (anchors: $+13.9$ vs. $+18.3$ pp semantic), step-level sign agreement is only 65%, and we make no claims about frontier or long-CoT reasoning models. The plan's nominal second generator (DeepSeek-R1-Distill) could not complete the protocol (Section 4); substituting phi-4 was a forced deviation, disclosed rather than hidden. **PRM coverage.** The deployed-PRM audit covers one trained checkpoint (Qwen2.5-Math-PRM-7B) plus two *prompted* LLM-judge PRMs, which approximate but do not equal trained generative PRMs (Khalifa et al., 2025); Math-Shepherd-style potential rewards (Wang et al., 2024) are not covered. **Semantic label reliability.** Step types come from AI-assisted annotation with human calibration; the two label passes disagree on 27.3% of steps. A blinded pilot (120 steps, two independent human labels each, four annotators in two pairs) reached only 61.7% raw agreement ($\kappa = 0.40$), concentrated on the essential/redundant boundary; where annotators agreed, their consensus tracked the frozen outcomes (93% of consensus-Harmful steps carry rating -1; pure semantic effect $+22.9$ pp). A third annotator adjudicated all 46 disagreements, yielding final labels for all 120 pilot steps with 93% adjudication accuracy. Type-based subgroup boundaries are therefore approximate; the headline interaction survives under either label set. **Judge noise.** The 93.5%-agreement human audit and ≤ 0.13 pp substitution bound cover Qwen3-8B outputs; phi-4 and ProcessBench continuations are scored by the same audited judge pipeline but without a fresh per-generator human audit. **Placebo scope.** Matched placebos exist for 511/600 primary steps (position SMD ≤ 0.298); the external cohort's placebo matching uses character length ($\pm 20\%$) rather than tokenizer length, and 96/300 external steps admit no match. **Offline policy evaluation.** Risk–coverage and rollback numbers reuse frozen runs; an online system would face distribution shift from its own deletions. The rollback probe is drawn from the same run pool it is evaluated against, which can flatter acceptance quality slightly. **Paraphrase validity.** C3 paraphrases passed automatic meaning/length checks (7.9% excluded), and a blinded 140-item human fidelity audit rated 97.1% faithful or minimally deviating, with zero meaning changes among the 46 audited anchor-group paraphrases; two neutral-group paraphrases (4.3% of that group's audited items) were judged meaning-changed. C3 conclusions rest on this audited premise.

Ethics Statement

The study uses the public PRM800K dataset (MIT-licensed problems from MATH) and open-weight models; no human subjects beyond author annotation were involved, and no personal data is processed. Deleting reasoning steps is an accuracy/efficiency intervention; our safety framing (harm budgets) is about answer correctness, not user harm. We release artifacts to support reproduction and auditing.

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A Judge Audit Detail

Audited outputs	200
Binary agreement	93.5%
Correct-class F1	0.931
Pair-transition agreement (60 pairs)	91.7%
Max condition bias	2.3 pp
Δ overall target effect after substitution	-0.13 pp
Δ pure semantic effect after substitution	-0.05 pp

Table 7: Judge audit summary. Gates (agreement $\geq$ 90%, bias $\leq$ 5 pp) pass; all headline numbers are insensitive to full substitution of adjudicated labels.

B Reproducibility

Primary numbers derive from two frozen master tables (600 steps; 6,314 runs) via a dependency-free Python pipeline with fixed seeds (cluster bootstrap seed 20260722, downstream 20260723); a single command regenerates every table, figure source, and the numeric manifest consumed by this paper. The extension cohorts add 6,965 runs (cross-generator 2,691; strong controls 1,862; ProcessBench 2,412), each with its own checkpointed generation, judging, and analysis script under the corresponding workstream directory. The release includes the intervention builder, matched-placebo selector, judge-audit tooling, policy risk–coverage and rollback code, the 120-step blinded annotation pilot, PRM-scoring adapters, the M5 signal-extraction script, and the StepRem benchmark split with its evaluation script.